

Java Assignment Questions

Assignment 1: Java Basics

1. Write a Java program to print 'Hello, World!'.
2. Perform arithmetic operations.
3. Swap two numbers without using a third variable.
4. Check whether a number is even or odd.
5. Find the largest of three numbers.

Assignment 2: Control Statements

1. Check whether a year is a leap year.
2. Display the multiplication table of a given number.
3. Find the factorial of a number.
4. Generate the Fibonacci series.
5. Check whether a number is prime.

Assignment 3: Arrays

1. Find the largest and smallest element in an array.
2. Sort an array in ascending order.
3. Search an element using linear search.
4. Search an element using binary search.
5. Find the second largest element in an array.

Assignment 4: Strings

1. Check whether a string is a palindrome.
2. Count vowels, consonants, digits, and special characters.
3. Reverse a string.
4. Check whether two strings are anagrams.
5. Count the frequency of each character.

Assignment 5: Methods

1. Create a method to calculate the square of a number.
2. Write a method to check whether a number is prime.
3. Demonstrate method overloading.
4. Write a recursive method to calculate factorial.
5. Write a recursive method for Fibonacci numbers.

Assignment 6: OOP

1. Create a Student class with data members and methods.

2. Demonstrate constructors.
3. Demonstrate inheritance.
4. Demonstrate method overriding.
5. Demonstrate encapsulation.

Assignment 7: Interfaces and Abstract Classes

1. Create an abstract Shape class and implement Circle and Rectangle.
2. Create a Vehicle interface and implement Car and Bike.
3. Demonstrate multiple inheritance using interfaces.
4. Implement runtime polymorphism.
5. Demonstrate abstraction.

Assignment 8: Exception Handling

1. Handle division by zero using try-catch.
2. Demonstrate multiple catch blocks.
3. Create a user-defined exception.
4. Use throw and throws.
5. Demonstrate the finally block.

Assignment 9: File Handling

1. Write data to a text file.
2. Read data from a text file.
3. Copy contents from one file to another.
4. Count words and lines in a file.
5. Append data to an existing file.

Assignment 10: Collections

1. Store student names using ArrayList.
2. Remove duplicate elements using HashSet.
3. Store key-value pairs using HashMap.
4. Sort elements using Collections.sort().
5. Iterate through collections using Iterator.

Assignment 11: Multithreading

1. Create a thread using the Thread class.
2. Create a thread using the Runnable interface.
3. Demonstrate thread synchronization.
4. Demonstrate thread priorities.
5. Use sleep() and join() methods.

Mini Projects

1. Student Management System
2. Library Management System
3. Bank Management System
4. Employee Payroll System
5. Hotel Reservation System
6. Online Shopping Cart
7. ATM Simulation
8. Quiz Application
9. Inventory Management System
10. Hospital Management System